

Safe C++ @ O'Reilly

Klaus Iglberger
August, 14th, 2026

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1. Introduction

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Content

1. Introduction of the Trainer
2. Motivation
3. Scope of the Training
4. Content of the Training
5. Schedule
6. Miscellaneous
7. Guidelines
8. Programming Tasks

Introduction of the Trainer

C++ Trainer/Consultant since 2016

Author of “C++ Software Design”

(Co-)Organizer of the Munich C++ user group

(Co-)Chair of the Software Design track at CppCon

Regular presenter at C++ conferences



Klaus Iglberger

Scope of this Training

In this training I will address ...

- ... the perceived safety problem of C++
- ... a discussion of possible solutions for those safety problems
- ... the philosophy of “Modern C++”

Content of this Training

1. Introduction
2. Safe C++
3. Summary/Literature

Schedule

Friday, August, 14th, 2026 (EDT time zone)

10:00am - 10:10am	<i>Introduction</i>
10:10am - 10:30am	<i>Motivation: The Many Safety Problems of C++</i>
10:30am - 11:00am	<i>Resolving Undefined Behavior with constexpr</i>
11:00am - 11:10am	<i>Break</i>
11:10am - 11:40pm	<i>Resolving Out-Of-Bounds Problems with Ranges</i>
11:40am - 12:00pm	<i>Resolving Type Safety with Strong Types (I)</i>
12:00pm - 12:10pm	<i>Break</i>
12:10pm - 12:30pm	<i>Resolving Type Safety with Strong Types (II)</i>
12:30pm - 1:00pm	<i>Resolving Memory Safety with Value Semantics (I)</i>
1:00pm - 1:10pm	<i>Break</i>
1:10pm - 1:30pm	<i>Resolving Memory Safety with Value Semantics (II)</i>
1:30pm - 1:55pm	<i>Dealing with the Sharp Edges</i>
1:55pm - 2:00pm	<i>Summary</i>

Miscellaneous

I assume you have some experience with C++. If something remains unclear or somewhat vague, **please ask!**

Miscellaneous

Also remember: **You are in control!**

Guidelines

(Core) Guideline: ...

The green boxes represent ...

- ... the most important take-aways;
- ... common idioms and best practice;
- ... markers in the slides.

Programming Tasks

Task (Subchapter/Name): ...

The yellow boxes represent ...

- ... hands-on programming tasks;
- ... links to the provided source code examples.

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